Modular arithmetic

L3-NT1 Level 3 Senior · Number theory

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November 2026

Use congruences, inverses, Fermat's little theorem and the Chinese remainder theorem to find remainders of very large numbers.
By the end you can compute remainders of huge powers using Fermat's little theorem and cycles; solve linear congruences by finding and using inverses; split a composite modulus and recombine with the Chinese remainder theorem.
Prerequisites: Factors, primes and divisibility (L2-NT1), Remainders and last digits (L2-NT2).
Contests: UKMT SMC, Senior Kangaroo, Maclaurin, AMC 12, AIME. **Time:** about 90 minutes.

1 Lesson

Consider the last two digits of 7^{2026}, a number with 1713 digits. Note that $7^2 = 49$ and $7^4 = 2401$. The last two digits of 7^4 are 01, so multiplying by 7^4 never changes the last two digits of anything. Since $2026 = 4 \cdot 506 + 2$, we can write $7^{2026} = (7^4)^{506} \cdot 7^2$, and this ends in the same two digits as $1 \cdot 49$. The answer is 49.

Every step of that argument uses only remainders, and modular arithmetic is the notation that justifies discarding the rest of the number. This unit develops modular arithmetic through congruences, inverses, Fermat's little theorem and the Chinese remainder theorem.

1.1 Congruences

We write $a \equiv b \pmod{n}$, read "a is congruent to b modulo n", when n divides $a - b$. Equivalently, a and b leave the same remainder on division by n. So $38 \equiv 3 \pmod{7}$, and also $38 \equiv -4 \pmod{7}$. Every integer is congruent to exactly one of $0, 1, \ldots, n-1$, but negative representatives are often the most useful ones. For example, $99 \equiv -1 \pmod{100}$, and $2^{10} = 1024 \equiv -1 \pmod{25}$.

Rules.
If $a \equiv b$ and $c \equiv d \pmod{n}$, then $a + c \equiv b + d$, $a - c \equiv b - d$ and $ac \equiv bd \pmod{n}$. Hence $a^k \equiv b^k \pmod{n}$ for every positive integer k.

The first two are immediate. For the product, write $ac - bd = a(c - d) + d(a - b)$, where both terms on the right are multiples of n. The power rule follows by using the product rule k times. In practice this means that in any sum, product or power you may replace a number by anything congruent to that number, at any stage. This is what we did with $7^4 \equiv 1 \pmod{100}$.

What you may *not* do in general is divide. We have $2 \cdot 3 \equiv 2 \cdot 8 \pmod{10}$, but $3 \not\equiv 8 \pmod{10}$. Division needs the next idea.

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1.2 Inverses and linear congruences

An *inverse* of a modulo n is a number a' with $aa' \equiv 1 \pmod{n}$. For example $3 \cdot 5 = 15 \equiv 1 \pmod{7}$, so 5 is an inverse of 3 modulo 7. If a has an inverse, then "dividing by a" means multiplying by a'.

Theorem.
The integer a has an inverse modulo n if and only if $\gcd(a, n) = 1$. In that case, for every b, the congruence $ax \equiv b \pmod{n}$ has exactly one solution modulo n.

Proof. Suppose $d = \gcd(a, n) > 1$. If $ax \equiv 1 \pmod{n}$, then $ax - 1 = kn$ for some integer k. But d divides both ax and kn, so d divides 1, which is absurd. Conversely, suppose $\gcd(a, n) = 1$ and look at the n numbers $0, a, 2a, \ldots, (n-1)a$. If two of them were congruent, say $ia \equiv ja$ with $0 \le j < i \le n - 1$, then n would divide $(i - j)a$. As n shares no prime factor with a, unique factorisation forces $n \mid i - j$, which is impossible since $0 < i - j < n$. So these n numbers leave n different remainders, which means every remainder appears exactly once. In particular one of them is $\equiv 1$, and exactly one is $\equiv b$. ■

In a contest you rarely need Euclid's algorithm to find an inverse. Try small multiples, or better, look for a multiple of a close to a multiple of n. Modulo 100, for instance, $33 \cdot 3 = 99 \equiv -1$, so $33 \cdot (-3) \equiv 1$ and the inverse of 33 is $-3 \equiv 97$.

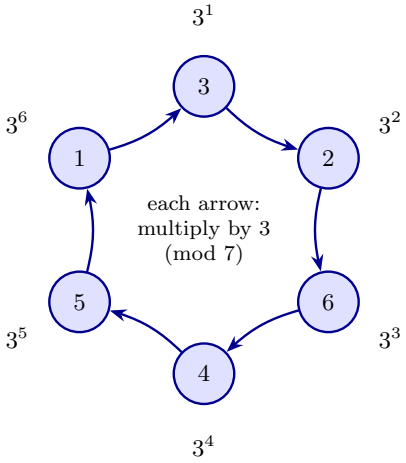
Example 1.
Find the smallest positive integer n such that $17n$ leaves remainder 5 when divided by 23.

Solution.
We need $17n \equiv 5 \pmod{23}$. Testing $n = 1, 2, 3, \ldots$ happens to be quick here, but we use a method that also works for large moduli. First make the coefficients small. We have $17 \equiv -6$, so the congruence reads $-6n \equiv 5$, that is $6n \equiv -5 \pmod{23}$.
Now we want the right-hand side to be a multiple of 6, so keep adding 23 to get $-5 \equiv 18 \pmod{23}$, and $18 = 6 \cdot 3$. Since $\gcd(6, 23) = 1$, the number 6 is invertible modulo 23, so we may cancel 6 from $6n \equiv 6 \cdot 3$ and get $n \equiv 3 \pmod{23}$. The smallest positive answer is $n = 3$, and indeed $17 \cdot 3 = 51 = 2 \cdot 23 + 5$.
The method is to replace numbers by convenient representatives (17 by -6, then -5 by 18) until a common factor can be cancelled. The cancellation was valid only because 6 and 23 are coprime.

1.3 Fermat's little theorem

Consider the powers of 3 modulo 7. They are $3, 2, 6, 4, 5, 1$, and then the pattern repeats. We reach 1 after $p - 1 = 6$ steps, and the theorem below says that $a^{p-1} \equiv 1$ for every a not divisible by p.

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Fermat's little theorem.
If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$. Multiplying by a, every integer a satisfies $a^p \equiv a \pmod{p}$.

Proof. By the proof of the theorem on inverses (with $n = p$), the numbers $0, a, 2a, \ldots, (p-1)a$ are pairwise incongruent modulo p. Only the first is $\equiv 0$, so $a, 2a, \ldots, (p-1)a$ are congruent to $1, 2, \ldots, p-1$ in some order. Multiplying them all together gives $a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}$. Each of $1, 2, \ldots, p-1$ is coprime to p, so $(p-1)!$ is coprime to p and can be cancelled, leaving $a^{p-1} \equiv 1$. ■

In practice this means that modulo a prime p, and for $p \nmid a$, exponents can be reduced modulo $p-1$. For instance, modulo 11 we have $a^{10} \equiv 1$, so $a^{2026} = (a^{10})^{202} \cdot a^6 \equiv a^6$. Sometimes the cycle is shorter than $p-1$ (the powers of 2 modulo 7 go $2, 4, 1$), but its length always divides $p-1$, as you will prove in L4-NT1. For a composite modulus like 100 there is no $p-1$ to use. Either find the cycle directly, as we did for 7, or split the modulus, as below.

Example 2.
Find the remainder when 3^{2026} is divided by 31.

Solution.
Since 31 is prime and does not divide 3, Fermat gives $3^{30} \equiv 1 \pmod{31}$. Now $2026 = 30 \cdot 67 + 16$, so $3^{2026} \equiv 3^{16}$. This is still a large power, so we look for a small power of 3 close to a multiple of 31. First $3^3 = 27 \equiv -4$, which is small but not obviously useful. Then $3^5 = 243 = 7 \cdot 31 + 26$, so $3^5 \equiv 26 \equiv -5$.
Now $3^{15} = (3^5)^3 \equiv (-5)^3 = -125$, and $125 = 4 \cdot 31 + 1$, so $-125 \equiv -1$. Hence $3^{15} \equiv -1$, and $3^{16} \equiv -3 \equiv 28 \pmod{31}$. The remainder is 28.
Note the two habits. Reduce the exponent first using Fermat, then keep the base small using negative representatives, so that no number in the calculation ever gets big.

1.4 The Chinese remainder theorem

Knowing a number modulo 4 and modulo 25 tells you the number modulo 100. More generally, we have the following.

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Chinese remainder theorem (CRT).
If $\gcd(m,n)=1$, then for any integers a and b the system $x\equiv a \pmod m$, $x\equiv b \pmod n$ has a solution, and any two solutions are congruent modulo mn.

Proof. The n numbers $a, a+m, a+2m, \dots, a+(n-1)m$ all satisfy the first congruence. If two of them were congruent modulo n, then n would divide $(i-j)m$ for some $0<i-j<n$, which is impossible since $\gcd(m,n)=1$, exactly as in the theorem on inverses. So they leave n different remainders modulo n, and exactly one of them is $\equiv b$. If x and y are both solutions, then $m \mid x-y$ and $n \mid x-y$, and since m and n are coprime, $mn \mid x-y$. ■

So each pair (remainder mod m, remainder mod n) corresponds to exactly one remainder mod mn. The grid below shows this for 3 and 5. Each of $0,1,\dots,14$ sits in its own cell, and every cell is used once. The shaded cell says that the only number in this range that is 2 mod 3 and 4 mod 5 is 14.

$n \bmod 3$ \ $n \bmod 5$	0	1	2	3	4
0	0	6	12	3	9
1	10	1	7	13	4
2	5	11	2	8	14

The theorem extends to any number of pairwise coprime moduli. The coprimality matters. The system $x\equiv 1 \pmod 4$, $x\equiv 2 \pmod 6$ has no solution at all, because the first condition makes x odd and the second makes x even.

CRT is how you handle composite moduli. To work modulo 100, work modulo 4 and modulo 25 separately, and for 1000 use 8 and 125. The pieces are usually much easier, especially when the base is not coprime to the full modulus.

Example 3.
Find the last two digits of 2^{2026}.

Solution.
One might look for a power of 2 that is $\equiv 1 \pmod{100}$, as we did with 7, but there is none. Every power 2^k with $k\geq 1$ is even, and an even number is never 1 more than a multiple of 100. The powers of 2 do eventually cycle modulo 100, but the cycle does not pass through 1, and finding the cycle by hand is slow.
So split $100=4\cdot 25$. Modulo 4 the question is trivial, since $2^{2026}\equiv 0$. Modulo 25, the number 2 is invertible, and $2^{10}=1024\equiv -1$. So $2^{20}\equiv 1 \pmod{25}$, and since $2026=20\cdot 101+6$, we get $2^{2026}\equiv 2^6=64\equiv 14 \pmod{25}$.
Now combine. We want $0\leq x<100$ with $x\equiv 14 \pmod{25}$ and $x\equiv 0 \pmod 4$. The first condition gives the candidates $14, 39, 64, 89$, and only 64 is a multiple of 4. So 2^{2026} ends in 64.

Example 4.
Find the remainder when 7^{7^7} is divided by 33. (The exponent is $7^7=823543$.)

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Solution.
The modulus is composite, so split $33 = 3 \cdot 11$. Modulo 3 there is nothing to do, since $7 \equiv 1$, so $7^{7^7} \equiv 1 \pmod 3$.
Modulo 11, Fermat gives $7^{10} \equiv 1$, so all we need is the exponent 7^7 modulo 10. This is the general method for towers. *To reduce the top exponent modulo $p-1$, you need a second, smaller modular calculation.* The last digits of the powers of 7 cycle $7, 9, 3, 1$ with period 4, and $7 = 4 + 3$, so 7^7 ends in 3. Therefore $7^7 = 10k + 3$ for some integer k, and

$$7^{7^7} = \left(7^{10}\right)^k \cdot 7^3 \equiv 7^3 = 343 \pmod{11}.$$

As $343 = 31 \cdot 11 + 2$, this is 2.
Finally solve $x \equiv 1 \pmod 3$ and $x \equiv 2 \pmod{11}$. The candidates for the second are $2, 13, 24$, and only 13 is 1 more than a multiple of 3. The remainder is 13.
The tempting mistake is to reduce the exponent 7^7 modulo 33, or modulo 11. Exponents are reduced modulo the length of the cycle, never modulo the original modulus.

Common traps.

- Cancelling a factor that shares a divisor with the modulus. From $6x \equiv 6 \pmod 9$ you may conclude only that $x \equiv 1 \pmod 3$, so $x \equiv 1, 4$ or $7 \pmod 9$.
- Reducing an exponent modulo p instead of $p-1$, or modulo n itself for a composite n.
- Using Fermat when $p \mid a$ (then $a^{p-1} \equiv 0$), or with a composite modulus, as in $2^8 = 256 \equiv 4 \pmod 9$, not 1.
- Applying CRT to moduli that are not coprime. Conditions mod 4 and mod 6 can contradict each other, and when they agree they pin x down only modulo 12.
- Leaving the answer as -3 when the question wants a remainder between 0 and $n-1$.

1.5 How to spot this in a contest

The usual signals are "the remainder when ... is divided by", "the last two (or three) digits", "the units digit", a huge exponent, a year number such as 2026 in an exponent, or "for which n is ... divisible by ...". When you see one, ask these questions in order.

1. Is the modulus prime? Then reduce exponents modulo $p-1$ using Fermat.
2. Is the modulus composite? Split the modulus into coprime prime powers and recombine with CRT, using 4 and 25 for 100, and 8 and 125 for 1000.
3. Is there a small power equal to ± 1? Hunting for ± 1, like $2^{10} \equiv -1 \pmod{25}$, is often faster than any theorem.
4. Is the exponent itself a power? Then compute the exponent modulo the cycle length first.

In multiple-choice papers, remember the cheap checks too. Work mod 9 with digit sums, mod 11 with alternating digit sums, and mod 10 with last digits. A single small modulus often eliminates three or four options before you do any real work.

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2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ Find the remainder when 2026^{2026} is divided by 7.

Problem 2 ⋆⋆ Find the smallest positive integer n such that the last two digits of $37n$ are 01.

Problem 3 ⋆⋆ A positive integer N leaves remainder 3 when divided by 7, remainder 4 when divided by 9, and remainder 5 when divided by 11. Find the smallest possible value of N.

Problem 4 ⋆⋆⋆ How many integers x with $1 \leq x \leq 1001$ have the property that $x^3 - x$ is divisible by 1001?

Problem 5 ⋆⋆⋆ Find the remainder when $2^{3^{2026}}$ is divided by 13. (The exponent is 3^{2026}.)

Problem 6 ⋆⋆⋆⋆ Find the remainder when $1^{2026} + 2^{2026} + 3^{2026} + \cdots + 2026^{2026}$ is divided by 13.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AMC 10B 2016 Problem 8** ⋆
   Only the last two digits matter, so work mod 100.
2. **AMC 12A 2008 Problem 15** ⋆⋆
   Units digits of powers cycle, so find the exponent modulo the cycle length.
3. **AIME I 2017 Problem 2** ⋆⋆
   Equal remainders means the divisor divides the differences.
4. **UKMT SMC 2026 Question 8** ⋆⋆
   Start from the largest candidate and track its remainder mod 7 as the candidate changes.
5. **BMO1 2019/20 Problem 1** ⋆⋆
   Look at all the numbers involved modulo a small prime.
6. **AMC 12B 2017 Problem 19** ⋆⋆⋆
   Split 45 into coprime factors, use digit tests, then recombine.
7. **AMC 10B 2011 Problem 23** ⋆⋆⋆
   Work mod 1000 and keep the base small.
8. **AIME 1983 Problem 6** ⋆⋆⋆
   Work mod 49 and write each base as 7 plus or minus 1.
9. **BMO1 2008/09 Problem 4** ⋆⋆⋆
   Reduce the numerator modulo the divisor by replacing n with a congruent value.
10. **AIME I 2010 Problem 2** ⋆⋆⋆
    Most of the factors are congruent to -1 mod 1000.

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11. **BMO1 2016/17 Problem 6** ⋆⋆⋆⋆

Turn the four divisibility conditions into statements about one common expression.

12. **AIME I 2011 Problem 11** ⋆⋆⋆⋆

Use CRT with 8 and 125, and find where the powers of 2 start repeating.

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4 Hints

1. Reduce the base modulo 7 first, then use Fermat to shrink the exponent.
2. You need $37n \equiv 1 \pmod{100}$. Look for a multiple of 37 that is very close to a multiple of 100.
3. Each remainder is a little more than half of its divisor. What can you say about $2N$?
4. Factorise 1001 into primes and factorise $x^3 - x$. Then count solutions modulo each prime and combine.
5. Fermat tells you to find the exponent modulo 12. Be careful that 3 is not coprime to 12, so split 12 into 4 and 3.
6. First reduce the exponent modulo 12. Then show that $S = 1^{10} + 2^{10} + \cdots + 12^{10}$ is divisible by 13 by comparing S with $2^{10}S$.

5 Solutions

Problem 1

Find the remainder when 2026^{2026} *is divided by* 7.

Since $2023 = 7 \cdot 289$, we have $2026 \equiv 3 \pmod{7}$, so we need 3^{2026} modulo 7. As 7 is prime and does not divide 3, Fermat gives $3^6 \equiv 1 \pmod{7}$. Now $2026 = 6 \cdot 337 + 4$, so $3^{2026} \equiv 3^4 = 81 = 11 \cdot 7 + 4$. The remainder is 4.

Answer: 4

Problem 2

Find the smallest positive integer n *such that the last two digits of* $37n$ *are* 01.

We need $37n \equiv 1 \pmod{100}$, so n is the inverse of 37 modulo 100, which exists and is unique modulo 100 because $\gcd(37, 100) = 1$. Note that $37 \cdot 27 = 999 \equiv -1 \pmod{100}$. Multiplying by -1, we get $37 \cdot (-27) \equiv 1$, so $n \equiv -27 \equiv 73 \pmod{100}$. The smallest positive value is $n = 73$, and indeed $37 \cdot 73 = 2701$.

Answer: 73

Problem 3

A positive integer N *leaves remainder* 3 *when divided by* 7, *remainder* 4 *when divided by* 9, *and remainder* 5 *when divided by* 11. *Find the smallest possible value of* N.

Doubling the three conditions gives $2N \equiv 6 \equiv -1 \pmod{7}$, $2N \equiv 8 \equiv -1 \pmod{9}$ and $2N \equiv 10 \equiv -1 \pmod{11}$. So $2N + 1$ is divisible by 7, 9 and 11, and since these are pairwise coprime, $2N + 1$ is divisible by $7 \cdot 9 \cdot 11 = 693$. Conversely, if $693 \mid 2N + 1$ then the three original conditions hold, because 2 is invertible modulo 7, 9 and 11 and the steps can be reversed. Now $2N + 1$ is odd, so $2N + 1$ is an odd multiple of 693, and the smallest is 693 itself, giving $N = 346$. We check that $346 = 49 \cdot 7 + 3 = 38 \cdot 9 + 4 = 31 \cdot 11 + 5$.

Answer: 346

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Problem 4

How many integers x with $1 \leq x \leq 1001$ have the property that $x^3 - x$ is divisible by 1001?

We have $1001 = 7 \cdot 11 \cdot 13$ and $x^3 - x = (x-1)x(x+1)$. For a prime p, the prime p divides this product if and only if p divides one of the three factors, that is, if and only if $x \equiv 0, 1$ or $-1 \pmod{p}$. For $p = 7, 11, 13$ these are three different residues, so there are exactly 3 good residues modulo each prime.

Since 7, 11 and 13 are pairwise coprime, $1001 \mid x^3 - x$ exactly when each of the three primes divides $x^3 - x$. By the Chinese remainder theorem, each choice of a good residue modulo 7, modulo 11 and modulo 13 corresponds to exactly one residue modulo 1001, so there are $3 \cdot 3 \cdot 3 = 27$ good residues modulo 1001. The integers $1, 2, \ldots, 1001$ contain each residue modulo 1001 exactly once, so the answer is 27.

Answer: 27

Problem 5

Find the remainder when $2^{3^{2026}}$ is divided by 13. (The exponent is 3^{2026}.)

Since 13 is prime, $2^{12} \equiv 1 \pmod{13}$, so we need 3^{2026} modulo 12. We cannot use Fermat-style reasoning modulo 12 directly, since 12 is not prime and 3 is not coprime to 12, so use the Chinese remainder theorem. Modulo 3, clearly $3^{2026} \equiv 0$. Modulo 4, $3 \equiv -1$ and 2026 is even, so $3^{2026} \equiv 1$. The residue modulo 12 that is 0 mod 3 and 1 mod 4 is 9, so $3^{2026} = 12k + 9$ for some integer k.

Hence $2^{3^{2026}} = (2^{12})^k \cdot 2^9 \equiv 2^9 = 512 \pmod{13}$, and $512 = 39 \cdot 13 + 5$. The remainder is 5.

Answer: 5

Problem 6

Find the remainder when $1^{2026} + 2^{2026} + 3^{2026} + \cdots + 2026^{2026}$ is divided by 13.

Terms with $13 \mid k$ contribute 0. For $13 \nmid k$, Fermat gives $k^{12} \equiv 1$, and $2026 = 12 \cdot 168 + 10$, so $k^{2026} \equiv k^{10} \pmod{13}$. Since a term depends only on k modulo 13, each block of thirteen consecutive values of k contributes $S = 1^{10} + 2^{10} + \cdots + 12^{10}$ modulo 13.

The hard step is to show that $13 \mid S$. As r runs through $1, 2, \ldots, 12$, the numbers $2r$ run through all the non-zero residues modulo 13 in some order (because 2 is invertible modulo 13). Therefore $\sum (2r)^{10} \equiv \sum r^{10}$, that is, $2^{10} S \equiv S$, so $(2^{10} - 1)S \equiv 0 \pmod{13}$. But $2^{10} = 1024 = 78 \cdot 13 + 10$, so $2^{10} - 1 \equiv 9$, which is invertible modulo 13. Hence $S \equiv 0$.

Now $2026 = 13 \cdot 155 + 11$. The values $k = 1, \ldots, 2015$ form 155 complete blocks and contribute 0. The remaining values $k = 2016, \ldots, 2026$ are congruent to $1, \ldots, 11$, so they contribute $1^{10} + \cdots + 11^{10} = S - 12^{10} \equiv 0 - (-1)^{10} = -1$. The remainder is $13 - 1 = 12$.

Answer: 12

The online version, with progress tracking, is at kevinvariant08.github.io/pathways.html.

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